

1. Executive Summary

- Product name
 - One-line vision (problem being solved)
 - Target users
 - High-level solution
 - Expected business impact (revenue, efficiency, scale)
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2. Problem Statement

- Clear definition of the problem
 - Who is affected
 - Current solutions and their limitations
 - Why this problem needs to be solved now
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3. Target Users and Personas

- Primary users
 - Secondary users
 - Pain points for each persona
 - Expected usage patterns
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4. Product Goals and Success Metrics

- Business goals (revenue, growth, adoption)
 - Product goals (engagement, retention)
 - AI-specific metrics:
 - Accuracy
 - Latency
 - Cost per request
 - User satisfaction
 - Define measurable KPIs
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5. Scope Definition

In Scope

- Core features
- AI capabilities
- Integrations

Out of Scope

- Clearly define exclusions to prevent scope creep
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6. System Overview (High-Level Architecture)

- Frontend
- Backend
- AI or LLM layer
- Database
- APIs
- Third-party services

Include:

- Data flow
 - Request flow
 - User journey
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7. Agentic AI Design

This section is critical for agent-based systems.

Agent Definition

- Role of the agent
- Capabilities

Tools and Actions

- APIs it can call
- External systems it interacts with

Memory Design

- Short-term memory (session-based)
- Long-term memory (vector database or persistent store)

Reasoning Flow

- Decision-making logic
- Planning versus reactive behavior

Guardrails

- Prompt injection protection

- PII filtering
 - Output validation
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8. Feature Breakdown (Functional Requirements)

For each feature:

- Feature name
 - Description
 - User story (As a user, I want...)
 - Acceptance criteria
 - Edge cases
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9. API and Integration Requirements

- Internal APIs
 - External APIs (payments, messaging, video, etc.)
 - Data contracts
 - Rate limits
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10. Data and Storage Design

- Data sources
 - High-level database schema
 - Vector database (if applicable)
 - Data privacy considerations
 - Logging and tracing strategy
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11. Non-Functional Requirements

Performance

- Latency targets
- Throughput

Scalability

- Concurrent users
- Auto-scaling approach

Reliability

- Uptime expectations (SLA)

Security

- Authentication
- Authorization
- Encryption

Cost Constraints

- AI cost per request
 - Infrastructure cost
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12. AI Safety and Governance

- Bias mitigation strategy
 - Hallucination handling
 - Moderation policies
 - Compliance requirements (GDPR or others)
 - Audit logging
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13. Monitoring and Observability

- Logging
- Metrics
- Distributed tracing
- Alerts

AI-specific:

- Prompt and response logs
 - Evaluation metrics
 - Drift detection
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14. Testing Strategy

- Unit testing
 - Integration testing
 - AI evaluation (automated and human evaluation)
 - Load testing
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15. Deployment and DevOps

- CI/CD pipeline
 - Environment setup (development, staging, production)
 - Rollback strategy
 - Feature flags
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16. UX and UI Requirements

- User flows
 - Wireframes (optional but recommended)
 - Accessibility considerations
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17. Timeline and Milestones

- Phase-wise delivery plan
 - MVP definition
 - Iteration roadmap
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18. Business Model

- Pricing strategy
 - Monetization model
 - Cost versus revenue expectations
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19. Risks and Assumptions

- Technical risks
 - AI-specific risks
 - Business risks
 - External dependencies
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20. Open Questions

- Unknowns that need clarification
- Areas requiring validation or research